Laurent Schroeder

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Birth Date: 12/05/2002, Nationality: Luxembourg

Interest and Experience with Autonomous Systems

- At Avientus, I led the end-to-end mission planning development for an autonomous last-mile delivery drone. I was responsible for configuring the Pixhawk flight controller, designing the mission management logic, and integrating all hardware interfaces on an NVIDIA Jetson AGX Orin. We successfully advanced to the next stage of the SPRIND *Fully Autonomous Flight 2.0 Challenge* and received an award of €325k.
- Completed a semester project titled Language-Guided Drone Control, developing a system to convert voice commands into drone velocity commands via imitation learning. Achieved 75% success in simulation and 62.22% in real-world tests, including generalization to unseen but semantically similar voice inputs.
- Completed my Bachelor's thesis on the open-source drone platform **Crazyflie 2.1**, where I developed a control algorithm that fuses data from a forward-facing **Time-of-Flight sensor** and an **ultrasonic sensor**. The fusion strategy, designed to leverage the complementary strengths of both sensors for enhanced obstacle detection and navigation, demonstrated **80% reliability** in indoor environments with complex obstacles such as glass.
- I took the **Planning and Decision Making for Autonomous Robots** course, where we studied a wide range of planning algorithms, including informed search methods like Greedy Best-First Search and A^*, dynamic programming techniques such as Value Iteration and Policy Iteration for decision-making under uncertainty, and sampling-based planners like RRT^*. We also explored learning-based approaches and advanced optimization methods such as Successive Convexification (SCvx). Most of these algorithms were implemented as part of hands-on projects during the course.

Education

Sep 2024 - Present:	**Master in Robotics, Systems, and Control**, *ETH Zürich*.
Sep 2021 - Sep 2024:	**Bachelor in Electrical Engineering and Information Technology**, *ETH Zürich*.
Sep 2014 - Jun 2021:	**Classical Secondary Education-Advanced Level**, Lycée de Garçons de Luxembourg.

Work Experiences

- **Full Time Avientus Robotics Engineer Intern** (Sep 2025 - Mar 2026): Responsible for developing local and global planners, as well as the mission manager for an autonomous hexacopter for the SPRIND challenge.
- **ARIS Electronics Engineer NAUTILUS** (Oct 2024 - Mar 2025): Responsible for regulating the flow and reflow between the tank and the bladder of the Unmanned Underwater Vehicle (UUV).
- **Tennis Coach** (Feb 2023 - Jun 2023): Training juniors weekly.
- **Student Trainee at the national railway company (CFL) in Luxembourg** (Jun 2022 - Jul 2022): Research on renewable energies.

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Academic Courseworks

Model Predictive Control (MPC)

- Implemented a hands-on project to regulate truck-cabin temperatures under input/state constraints and external disturbances
- Gained strong foundations in stability analysis, constraint satisfaction, tracking MPC, soft constraints, and robust MPC formulations

Advanced Model Predictive Control (AMPC)

- Completed all four hands-on exercises, ranging from nominal linear MPC design to deploying MPC as a safety filter for learning-based controllers
- Developed stochastic MPC strategies as well as learning based controllers to control uncertain systems

Recursive Estimation

- Studied Bayesian state estimation methods, including Kalman Filter, Extended Kalman Filter, and Particle Filter
- Applied recursive estimation methods for real-time state tracking and reconstruction under uncertainty

Teamwork Experiences

- **Avientus Robotics Engineer:** Developed a fully autonomous drone capable of searching for objects in an unknown environment in a team of 3 people.
- **Duckietown:** Developed a navigation pipeline for driving a duckiebot through a city with traffic lights.
- **Real-World Robotics:** Developed a hand from the hardware to the software such that it is able to do complex tasks such as teleoperation, in a group of 7 people.
- **ARIS NAUTILUS:** Collaborated with 40 peers to develop an underwater vehicle for climate research and potentially exploration of Europa's ocean.

Additional Skills

- **Programming:** C++, C, Python, Matlab/Simulink, LaTeX

Hackathons

- **Hack2026 ETHRC:** In a team of six, developed and integrated multiple interaction capabilities on the Unitree G1 robot within 48 hours, building on the decoupled whole-body controller. Implemented real-time human following, hand signal interpretation, and natural language command response. Additionally, developed motion-to-video generation using the GENMO model.

Conferences and Workshops

- **2024 IEEE ISSE:** Presented the $SuperBat$ paper on nano-UAV research as lead author.

Publications

L. Schroeder, H.Müller, T.Polonelli, M.Magno, and L.Benini, **"Superbat - advancing obstacle avoidance on nano-uavs by fusing ultrasonic and laser-based time-of-flight sensors"**, IEEE ISSE 2024, pp. 1–8

Awards

Best Project-Based Learning (PBL) Bachelor Thesis 2024: Obstacle Avoidance With Ultrasonic Sensors on Nano Drones.

Languages

Luxembourgish	Native Speaker
English	Full Professional Proficiency
French	Full Professional Proficiency
German	Full Professional Proficiency
Swedish	Elementary Proficiency

Other Interests

- Sports: Tennis (Competition level), Skiing, Running
- Miscellaneous: Drones, Arduino programming, Travel

- **Tools:** ROS, ROS2, Gazebo, Git, CAD Design, PX4, Docker